

FIN — Release 10.30

Research Programs P337–P350

Certified signed resources, carrier naturality,
operational compilation, and conditional physics interfaces

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30 July 2026

Publication — Preprint; Version 1.0.0

Certified envelope result. Exact rational Bernstein and algebraic-input interval arithmetic proves $\mathcal{N}_{\text{env}} \geq 0.4067063265581317$.

Full oscillatory result. For the twelve phase-bearing strict moments,

$$0.7072784747 \lesssim \mathcal{N}_{\text{osc}} \lesssim 0.7073599749$$

is the strong numerical enclosure. It is a classical signed-path resource, not a physical-negativity theorem.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Publication statement

This monograph reports all executed Programs P337–P350 and recommends Programs P351–P364. Exact proof, high-precision evaluation, synthetic apparatus design, conditional sector physics, dimensional calibration, and missing external evidence remain separate. The legacy/strict split, QW-2191, dimensional-source, role-transfer, laboratory, L_{total} , Standard-Model/gravity, and Theory-of-Everything boundaries are preserved.

Contents

Publication statement	i
Confidence convention	1
1 Scope and non-negotiable boundaries	2
1.1 Kernel split	2
1.2 Ontology	2
1.3 Frozen open boundaries	2
2 Executive scientific findings	4
3 P337 — exact rational certification of the P323 dual	5
3.1 Problem	5
3.2 Rational witness	5
3.3 Algebraic input intervals	6
3.4 What is and is not closed	6
4 P338 — the full oscillatory signed path resource	7
4.1 Why the envelope is not the kernel	7
4.2 Continuum dual	7
4.3 Fixed-support primal	8
4.4 Interpretation boundary	8
5 P339 — continuous refreeze sensitivity	9
5.1 Exact reproduction of the old objective	9
5.2 Continuous bounded search	9
5.3 Hold-out obstruction	10
6 P340 and P345 — carrier naturality	11
6.1 Scalar completion theorem	11
6.2 C12 interpolation	11
6.3 Formula-level versus operator-level invariance	12

7 P341 — compiled photonic frame and inverse calibration	13
7.1 Ideal compilation	13
7.2 Short-time inverse calibration	13
7.3 Exact branch nonuniqueness	14
8 P343 — parallel channel-comb subclass	15
9 P344 — nonparametric clock equivalence	17
10 P346 — typed bridge-resource theory	19
10.1 Why one scalar resource is not justified	19
10.2 Signed path monotone	19
10.3 Other typed monotones	19
11 P347 — attempted resource co-generation	21
11.1 Minimum-phase/Kramers–Kronig candidate	21
11.2 Fractional-power continuation	21
11.3 Complex signed path representation	21
11.4 Abstract resource product	21
12 P348 — conditional electroweak falsification vector	23
13 P349 — conversion metrology	24
14 P342 and P350 — external admission gates	26
14.1 P342 hold-out	26
14.2 P350 hardware/reservoir	26
15 New theoretical objects O88–O100	27
16 Cross-program synthesis	28
16.1 The most important positive result	28
16.2 The most important negative result	28
16.3 The most useful missing-object construction	28
17 Recommended Research Programs P351–P364	29
17.1 Recommended immediate sequence	30
18 Reproduction	31
19 Final verdict	32

20 References and repository sources	33
Reproduction certificate	34

Confidence convention

Every substantive claim uses one of the following labels.

Label	Meaning in this report
[Proven]	Exact mathematical argument, exact finite calculation, or compiled formal structural theorem
[Strong evidence]	Stable high-precision or exhaustive finite computation with explicit numerical boundary
[Moderate evidence]	Reproducible calculation whose interpretation depends materially on modelling choices
[Speculative]	A research proposal without a completed certificate
[Refuted]	The declared claim or construction fails in its stated scope
[Blocked by external evidence]	Completion requires independent data, custody, calibration, hardware, or unblinding
[Conditional]	The conclusion follows only after named additional axioms or calibration standards are supplied

Chapter 1

Scope and non-negotiable boundaries

1.1 Kernel split

The two kernels remain distinct typed objects:

$$K_{\text{legacy,ont}}(d) = \alpha_{\text{geo}} \frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d}, \quad \alpha_{\text{geo}} = 4 \ln 2,$$

and

$$K_{\text{strict,gate}}(d) = \frac{\cos((743/4000)d + 13/80)}{1 + d^{9/5}}.$$

The legacy object is retained as an intermediate bridge kernel. The strict object is the later enriched gate-selected working kernel. This report does not silently identify them and does not transfer the historical electroweak, electromagnetic, or gravity roles.

1.2 Ontology

The repository ontology is preserved:

$$\text{informational nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer}.$$

The nadsoliton itself is the primordial information in a solitonic state. There is no separate information layer below it. The mathematical results below neither prove nor disprove that ontology; they classify what the current operator data can generate.

1.3 Frozen open boundaries

The following remain open:

- the non-premise selector and QW-2191;
- an internal dimensional source;
- a source-independent legacy-to-strict completion law;
- a legacy-role transfer theorem;
- independent laboratory validation;
- a unit-bearing, nonproxy L_{total} ;

- Standard Model or gravity generation;
- Theory-of-Everything closure.

Chapter 2

Executive scientific findings

Program	Main result	Status
P337	Exact rational dual lower certificate 0.4067063265581317	[Proven] lower bound
P338	Full oscillatory resource in [0.7072784747, 0.7073599749] numerically	[Strong evidence]
P339	Continuous best candidate improves score, but no hold-out exists	[Strong evidence] / [Refuted] hold-out
P340	C12-trained scalar map fails all four tested noncycle carriers	[Proven] necessary theorem / [Strong evidence] matrix
P341	66-rotation ideal mesh; single-time generator inversion is nonunique	[Strong evidence] / [Proven]
P342	No independent frozen-holdout manifest ad- mitted	[Blocked]
P343	Parallel n -use discrimination ladder computed for $n = 1, \dots, 4$	[Proven] subclass
P344	Finite times do not identify bounded-curvature monotone clocks	[Proven]
P345	Formula coefficients are carrier-independent inputs; spectra are not	[Proven] distinction
P346	Five typed resource monotones/free-operation classes	[Proven]
P347	Four candidate resource-coupling laws fail	[Refuted] in declared classes
P348	Five-component electroweak vector, condi- tional on sector physics	[Conditional]
P349	Rank-three $(\ell_*, \tau_*, \hbar_*)$ conversion frame	[Proven] algebra / [Conditional] standards
P350	No certified physical-reservoir package admit- ted	[Blocked]

Chapter 3

P337 — exact rational certification of the P323 dual

3.1 Problem

P323 found a matching high-precision primal/dual solution for

$$h_n = \frac{1}{1 + n^{9/5}}, \quad n = 0, \dots, 6.$$

The structural theorem was exact, but the displayed FIN values were evaluated numerically. P337 does not rerun a moment grid. It certifies a nearby rational dual polynomial.

3.2 Rational witness

Let $q_*(x)$ denote the cubic P323 extremal polynomial normalized by $q_*(0) = 1$. P337 rounds its coefficients rationally and contracts the result by

$$\rho = \frac{99\,999\,999}{100\,000\,000}.$$

Define

$$q_r(x) = \rho \tilde{q}(x), \quad p_r(x) = -q_r(x)^2.$$

Nonpositivity is exact because p_r is minus a rational square. P337 subdivides $[0, 1]$ into 512 rational intervals, converts q_r^2 on each interval to the Bernstein basis, and obtains the exact global upper bound

$$q_r(x)^2 \leq \frac{99999998000000001}{100000000000000000} = 0.99999998000000001 < 1.$$

Therefore

$$-1 < p_r(x) \leq 0$$

on the complete continuum.

3.3 Algebraic input intervals

For every integer $n = 1, \dots, 6$, P337 brackets $n^{1/5}$ between rationals a_n, b_n and verifies

$$a_n^5 \leq n \leq b_n^5$$

by exact integer arithmetic. Monotonicity gives rational enclosures for $n^{9/5}$ and h_n . Propagating coefficient signs through these intervals produces the exact weak-dual lower bound

$$\mathcal{N}_{\text{env}} \geq 0.4067063265581317.$$

The input-interval contribution has width below 5.94×10^{-22} .

3.4 What is and is not closed

The distance to the P323 100-digit primal evaluation is

$$8.1341 \times 10^{-9}.$$

Thus the value is essentially fixed for practical work. Nevertheless:

- the dual lower certificate is [\[Proven\]](#);
- the matching primal value is [\[Strong evidence\]](#);
- a formal interval-Newton proof of the three nodes, their weights, and all moment equalities remains P351.

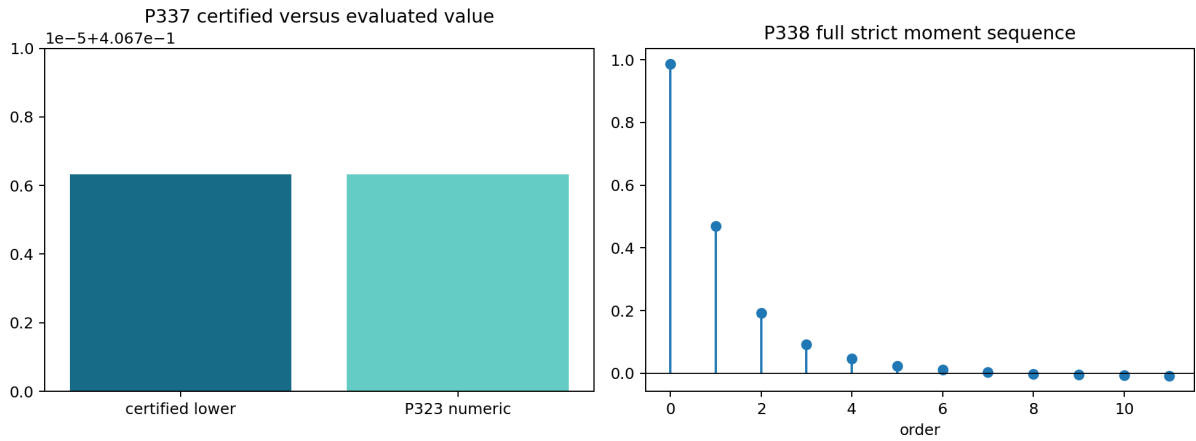


Figure 3.1: The exact rational envelope lower certificate and the full phase-bearing strict moment sequence.

Chapter 4

P338 — the full oscillatory signed path resource

4.1 Why the envelope is not the kernel

The P323 object used only the positive attenuation profile. The complete strict sequence contains the frozen phase:

$$k_n = \frac{\cos(\omega n + \phi)}{1 + n^\eta}, \quad \omega = \frac{743}{4000}, \quad \phi = \frac{13}{80}, \quad \eta = \frac{9}{5}.$$

For $n = 0, \dots, 11$, the moments k_8, k_9, k_{10}, k_{11} are negative. Therefore a signed representation of the full kernel is a genuinely different problem.

4.2 Continuum dual

For a signed measure $\mu = \mu_+ - \mu_-$ on $[0, 1]$, define

$$\mathcal{N}(\mu) = \mu_-([0, 1])$$

and impose

$$\int_0^1 x^n d\mu(x) = k_n, \quad n = 0, \dots, 11.$$

The dual searches degree-11 polynomials satisfying

$$-1 \leq p(x) \leq 0$$

and maximizes $\sum_n p_n k_n$. A cutting-plane calculation followed by global critical-point normalization yields

$$\mathcal{N}_{\text{osc}} \gtrsim 0.7072784746728757.$$

The reported polynomial range is

$$-0.9999999988 \leq p(x) \leq -9.97 \times 10^{-10}$$

under the numerical critical-point calculation.

4.3 Fixed-support primal

The raw monomial LP objective 0.7072626661 is **not** treated as an upper certificate: its tiny high-order equality residual is amplified by the ill-conditioned monomial basis. The LP is used only to select 12 rational support points. P338 then resolves all 12 moment equations at 100 digits on that fixed support. The residual is 1.43×10^{-101} , and the resulting negative mass is

$$\mathcal{N}_{\text{osc}} \lesssim 0.7073599748862595.$$

Consequently,

$$\boxed{0.7072784747 \lesssim \mathcal{N}_{\text{osc}} \lesssim 0.7073599749}$$

with width 8.15×10^{-5} . This is [\[Strong evidence\]](#), not yet an interval theorem.

4.4 Interpretation boundary

The difference

$$\mathcal{N}_{\text{osc}} - \mathcal{N}_{\text{env}} \gtrsim 0.30057214$$

is a classical signed Hausdorff-moment resource. It proves that the phase cannot be ignored when path representations are discussed. It does not select among:

- classical signed measures;
- quasiprobability;
- coherent interference;
- active feedback;
- an open-system memory kernel;
- information loss;
- negative information.

Chapter 5

P339 — continuous refreeze sensitivity

5.1 Exact reproduction of the old objective

P339 reuses the QW-2038 definitions of:

- the mass metric and its upstream fitted coefficients;
- CKM and PMNS reference matrices and flavor Hamiltonians;
- the GW feature table, score, κ , and thresholds;
- the original bounded parameter box.

At the frozen tuple

$$(\omega, \phi, \beta, \eta) = (0.18575, 0.16250, 1, 1.8),$$

the reproduced score is 1.8803719645, and all eight original gates pass.

5.2 Continuous bounded search

The best candidate found is

$$(\omega, \phi, \beta, \eta) \approx (0.18425313, 0.16250, 0.99999977, 1.79984890),$$

with score 1.8331321240. It also passes all eight original gates.

This is not called a global optimum: differential evolution reached its iteration cap. It is sufficient to prove that the selected grid point was not the best point found in the continuous box.

A 256-point local perturbation audit gives:

- pass fraction: 0.6836;
- score 5% quantile: 1.8728;
- median: 2.2319;
- score 95% quantile: 3.7398.

The frozen point lies in a nontrivial pass region, but it is neither isolated nor uniquely selected by the continuous objective.

5.3 Hold-out obstruction

No independent hold-out exists in the QW-2038 execution. Search and scoring reuse:

- the same particle masses;
- the same CKM and PMNS references;
- the same GW feature table;
- the same upstream fitted coefficients;
- the same thresholds.

There is no independent provider/registrar/analyst split, frozen hash, unblinding record, or hold-out manifest. Therefore:

Continuous stability of QW-2038 is in-sample procedural evidence. It is not an internal physical source, an ontology theorem, a selector, or a prediction validated out of sample.

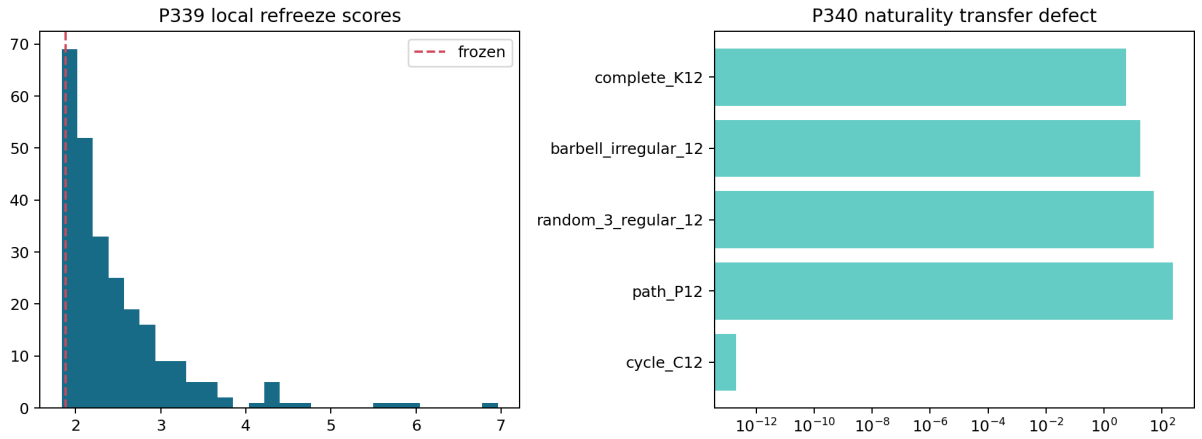


Figure 5.1: Continuous QW-2038 sensitivity and failure of the C12-trained scalar map on other carriers.

Chapter 6

P340 and P345 — carrier naturality

6.1 Scalar completion theorem

If

$$K_{\text{strict}} = f(K_{\text{legacy}})$$

by scalar functional calculus, then

$$[K_{\text{legacy}}, K_{\text{strict}}] = 0.$$

Commutation is necessary, not sufficient.

6.2 C12 interpolation

On C12 the two radial operators share the Fourier basis. Their seven distinct spectral classes define a degree-six polynomial f_{C12} with training residual 1.91×10^{-13} . This polynomial explicitly reads the strict C12 spectrum; it is a representation/interpolant, not a source theorem.

The identical polynomial is then transported without refitting:

Carrier	Relative transfer error	Relative commutator
C12	2.08×10^{-13}	5.20×10^{-17}
P12	232.61	0.1482
random 3-regular, 12 vertices	51.19	0.0919
irregular barbell	17.83	0.2605
K12	5.80	0

K12 is especially informative: it commutes, but the transfer still fails. Thus commutation cannot be promoted to completion.

6.3 Formula-level versus operator-level invariance

P345 repeats the strict radial formula on six carriers. The written constants

$$\omega = \frac{743}{4000}, \quad \phi = \frac{13}{80}, \quad \beta = 1, \quad \eta = \frac{9}{5}$$

are carrier-independent only because they are supplied to every carrier. Tested operator invariants vary:

Invariant	Coefficient of variation
normalized spectral entropy	0.0584
spectral radius per vertex	0.4652
Frobenius norm per $\sqrt{ E }$	0.0402

The natural object suggested by P340/P345 is not a universal scalar polynomial. It is a yet-unconstructed functorial completion law whose graph morphisms, carrier transport, and source data must be declared.

Chapter 7

P341 — compiled photonic frame and inverse calibration

7.1 Ideal compilation

At the P326 wave witness time

$$t_* = 1.645324972836305,$$

P341 compiles

$$U_* = \exp(-it_* A_{\text{strict}})$$

into:

- 66 complex two-mode Givens rotations;
- 12 terminal phases.

The ideal reconstruction residual is 1.07×10^{-15} .

7.2 Short-time inverse calibration

The generator is estimated at the calibrated short time $t_c = 0.08$:

$$\hat{A} = \frac{i}{t_c} \log \hat{U}(t_c),$$

after polar projection of the noisy matrix estimate. Under the synthetic complex matrix-noise model, the 95% witness-distribution TV errors are:

Complex noise SD	95% TV error at t_*
10^{-6}	3.90×10^{-5}
10^{-5}	4.47×10^{-4}
10^{-4}	4.15×10^{-3}
10^{-3}	3.98×10^{-2}

These are digital-transfer tolerances, not a hardware calibration.

7.3 Exact branch nonuniqueness

Let P be a spectral projector of A . Then

$$A' = A + \frac{2\pi}{t_*}P$$

satisfies

$$\exp(-it_*A') = \exp(-it_*A).$$

The constructed branch generator differs by operator norm $2\pi/t_* = 3.8188111$, while its unitary residual is 2.99×10^{-15} .

Therefore one-time generator tomography is mathematically nonidentifying. Short-time principal-log recovery is valid only after supplying a spectral bound and calibrated time.

Chapter 8

P343 — parallel channel-comb subclass

P343 declares:

- single-use space $\mathcal{H} = \mathbb{C}^{12}$;
- channels $\mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$;
- n -use parallel input $\mathcal{H}^{\otimes n}$;
- no intermediate memory;
- identity/no-intervention slots;
- an unrestricted final joint POVM.

For relative one-use phases z_j , the n -use relative spectrum contains all products

$$z_{j_1} \cdots z_{j_n}.$$

The distance of their convex hull from zero gives the ideal n -use unitary channel distance.

Parallel uses	First sampled perfect-discrimination time per use
1	0.72
2	0.36
3	0.24
4	0.18

The inverse- n pattern follows because repeated equal-time uses accumulate the same relative phase spread.

This is an exact parallel-unitary subclass. It is not an adaptive comb SDP: there is no memory system, causal instrument optimization, loss channel, or process tensor.

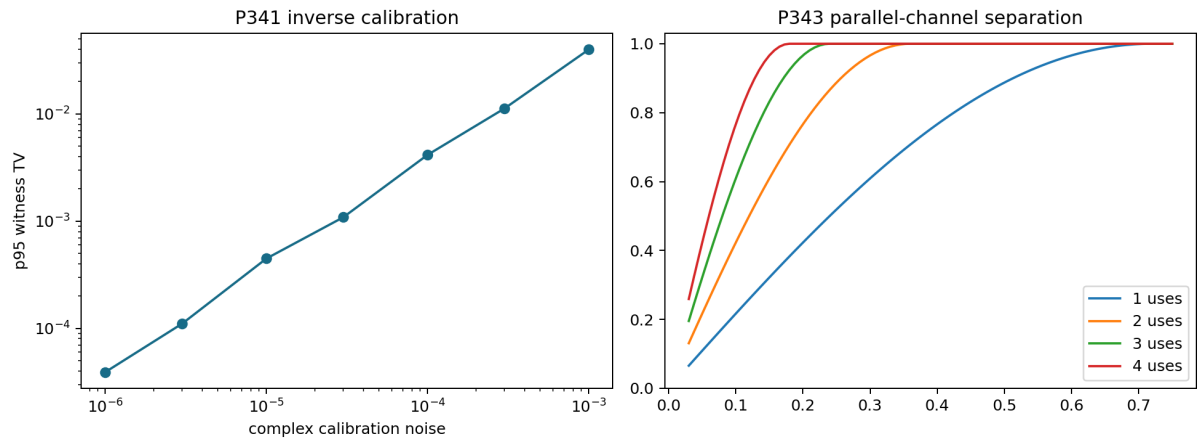


Figure 8.1: Synthetic short-time inverse calibration and ideal parallel-use channel discrimination.

Chapter 9

P344 — nonparametric clock equivalence

Consider the declared class

$$\tau(0) = 0, \quad |\tau''| \leq 0.2, \quad \tau' \geq 0.8.$$

For any largest unsampled interval $[a, b]$, define

$$\tau_{\pm}(t) = t \pm A \sin^2\left(\frac{\pi(t-a)}{b-a}\right), \quad A = \frac{M(b-a)^2}{2\pi^2},$$

inside the interval and use the identity clock outside it. The clocks and their first derivatives agree at the interval boundaries, belong to $W^{2,\infty}$, agree at every sampled time, remain monotone under the declared margin, and differ in the interior.

This proves finite-design nonidentification for the declared nonparametric clock class.

The P329 quartic Fisher design and the equispaced design give:

Design	Maximum gap	$Mh^2/8$ error bound	Two-clock midpoint separation
P329 quartic-Fisher design	0.57128	0.008159	0.006613
equispaced nonparametric design	0.28	0.001960	0.001589

The P329 design is good for its finite polynomial Fisher criterion. It is not minimax for a bounded-curvature nonparametric clock. Different uncertainty classes imply different optimal experiments.

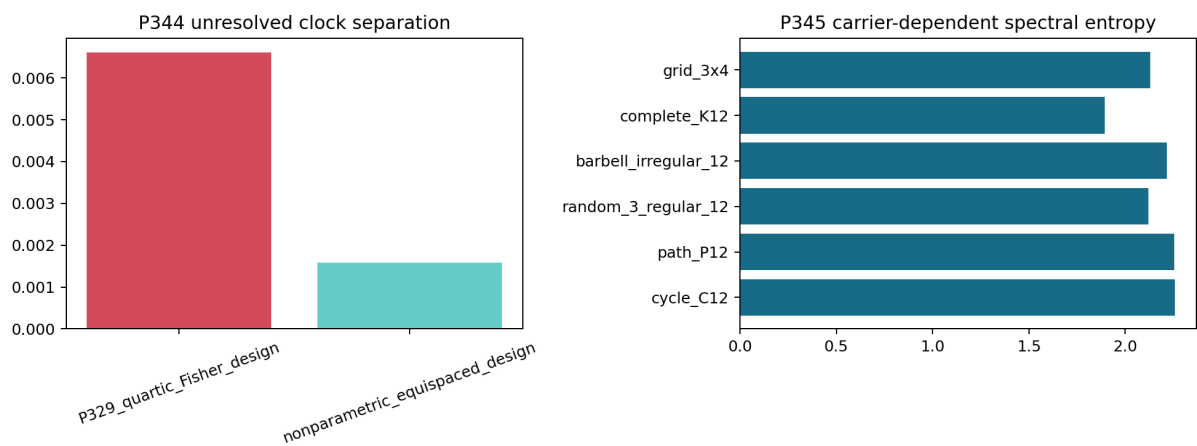


Figure 9.1: Nonparametric clock uncertainty and carrier-dependent operator invariants.

Chapter 10

P346 — typed bridge-resource theory

10.1 Why one scalar resource is not justified

P335 showed logical independence among:

1. signed path weight;
2. nontorsion phase;
3. an independent cross law;
4. pointing/orientation;
5. dimensional scale.

P346 defines a separate free-operation class and monotone for each resource. This is the smallest honest resource-theoretic object presently available:

$$\mathbf{M} = (\mathcal{N}_{\text{path}}, \text{Phase}_{\infty}, \text{Cross}, \text{Point}, \text{Scale}).$$

It is a typed vector, not a physically preferred scalar.

10.2 Signed path monotone

For a signed measure of fixed total mass,

$$\mathcal{N}(\mu) = \frac{\|\mu\|_{\text{TV}} - \mu(X)}{2}.$$

A positive mass-preserving Markov map T contracts total variation and preserves total mass. Therefore

$$\mathcal{N}(T\mu) \leq \mathcal{N}(\mu).$$

All 512 random finite tests satisfy the inequality. The median contraction ratio is 0.1492.

10.3 Other typed monotones

Resource	Free operation	Non-generation statement
nontorsion phase	group homomorphism from torsion in-	torsion cannot map to infinite order
put		
pointing	equivariant map without a section	a free transitive torsor does not cre-
		ate a point
positive scale	scale-free $\mathbb{R}_+$ -equivariant map	no invariant positive unit exists
cross law	deterministic legacy-input-only map	identical inputs cannot identify two
		distinct targets

The free-operation classes are mathematical choices. The report does not claim that FIN dynamics already realizes them.

Chapter 11

P347 — attempted resource co-generation

Four candidate laws were tested.

11.1 Minimum-phase/Kramers–Kronig candidate

If a modulus is the boundary value of a causal, analytic, zero-free transfer function, its phase can be reconstructed up to a constant and delay. Applying this counterfactually to the strict attenuation envelope requires adding:

- a causal frequency coordinate;
- analyticity;
- zero-freeness;
- boundary conditions.

Distance d is not a frequency variable in the current kernel. Even after granting these axioms, the discrete minimum-phase reconstruction has central phase RMSE 0.6727 after constant/delay freedom, far from the frozen linear phase. **[Refuted in the declared discretization]**

11.2 Fractional-power continuation

Analytic continuation of $z^{9/5}$ introduces branch phases, but they are branch-constant structures. They do not generate $\omega d + \phi$. **[Refuted structurally]**

11.3 Complex signed path representation

A complex measure can represent the full oscillatory moments, but only by reading those target moments. It is a representation, not a source-independent completion. **[Representation only]**

11.4 Abstract resource product

P335 single-omission models refute any logical implication among the five resources in the bare product signature. A coupling requires a new axiom or theorem. **[Refuted without a coupling law]**

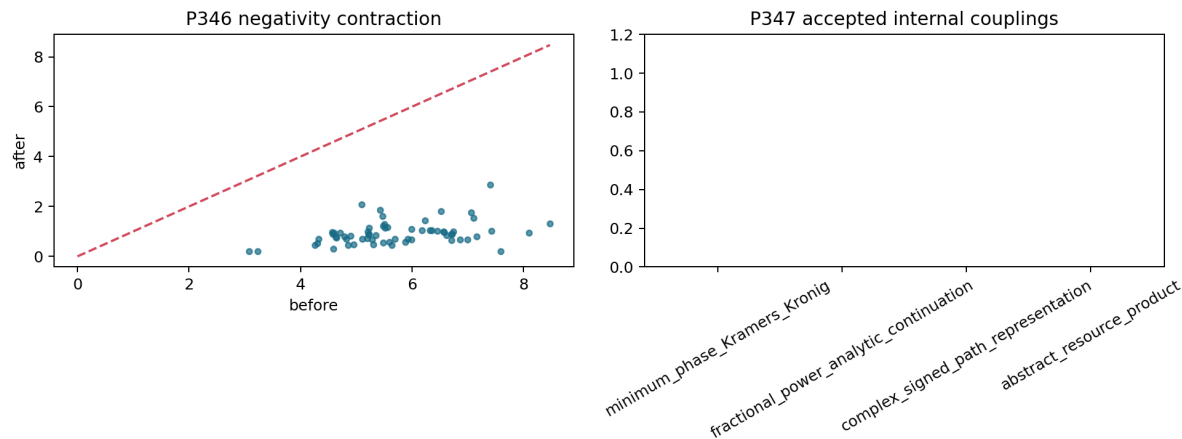


Figure 11.1: Signed-measure monotonicity under Markov maps and the tested resource-coupling matrix.

Chapter 12

P348 — conditional electroweak falsification vector

Let the historical role law be added explicitly:

$$x = \sin^2 \theta_W = \frac{\alpha_{\text{geo}}}{12} = \frac{\ln 2}{3} \approx 0.2310490602.$$

After separately supplying the Standard Model gauge representations and one-Higgs-doublet tree relations, one obtains:

Observable	Conditional value
$\sin^2 \theta_W$	0.2310490602
m_W/m_Z	0.8768984775
electron g_V/g_A	0.0758037593
electron g_L^2/g_R^2	1.3549950846
neutrino g_V/g_A	1

The Jacobian of this vector with respect to x has rank one. Thus these are not five independent FIN derivations. They are a one-parameter, jointly falsifiable consequence of a supplied physical sector.

A valid future test must freeze:

- the sector point;
- the role law;
- representations and charges;
- tree/radiative order;
- renormalization scheme and scale;
- the complete covariance-aware observable vector;
- one unblinding.

This report uses no current experimental values. No strict role transfer is claimed.

Chapter 13

P349 — conversion metrology

Introduce an explicit conversion frame

$$\text{CA} = (\ell_*, \tau_*, \hbar_*).$$

Then

$$t_{\text{phys}} = \tau_* t, \quad x_{\text{phys}} = \ell_* d, \quad S_{\text{phys}} = \hbar_* S,$$

$$E_* = \frac{\hbar_*}{\tau_*}, \quad H_{\text{phys}} = \frac{\hbar_*}{\tau_*} A, \quad m_* = \frac{\hbar_* \tau_*}{\ell_*^2}, \quad c_* = \frac{\ell_*}{\tau_*}.$$

For any calibration observable, the logarithmic dependence on CA is its dimension-exponent row. A collection identifies CA exactly when its exponent matrix has rank three.

P349 tests 20 calibration triples; 17 have rank three. The direct length/time/action standards form the identity matrix and have condition number one. Every dimensionless FIN observable has a zero log-unit Jacobian and therefore cannot identify CA.

This is the smallest clean operational bridge between a dimensionless generator and dimensional laboratory quantities. It is also an impossibility statement:

No collection of purely dimensionless FIN observables can determine $\ell_, \tau_*, \hbar_*$.*

The standards remain external. Setting $c_* = c$ or $\hbar_* = \hbar$ is calibration, not emergence.

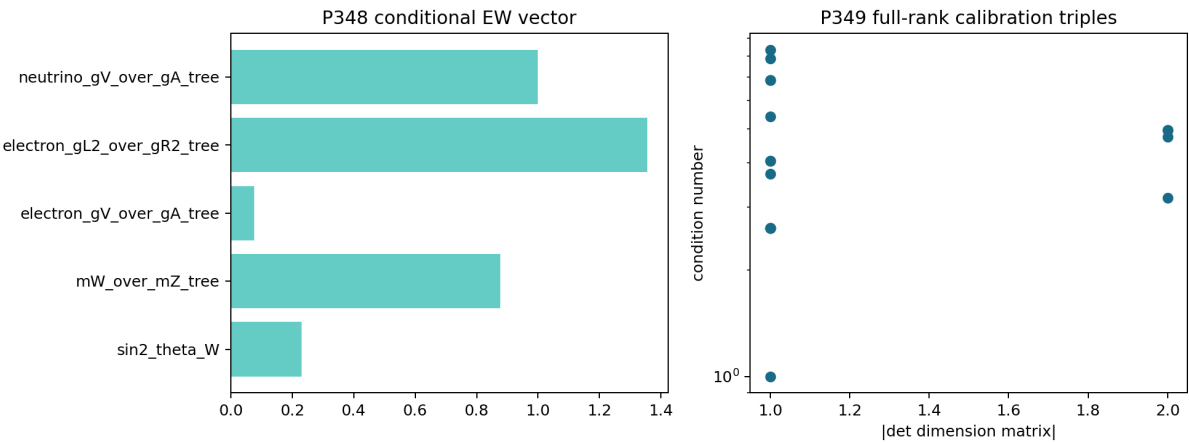


Figure 13.1: The conditional electroweak vector and rank-three dimensional conversion metrology.

Chapter 14

P342 and P350 — external admission gates

14.1 P342 hold-out

No manifest satisfies all of:

- independent provider, registrar, and analyst;
- frozen dataset hash;
- freeze timestamp;
- unblinding authorization.

Therefore an external refreeze validation is not authorized.

14.2 P350 hardware/reservoir

No admitted package supplies:

- a calibrated physical device or reservoir;
- raw event hashes;
- paired controls and seeds;
- frozen primary endpoint;
- distinct custody roles;
- one-shot execution authorization.

No hardware execution was performed.

Chapter 15

New theoretical objects O88–O100

Object	Definition	What it resolves	What it does not resolve
O88 Rational Bernstein Moment Certificate	exact $p_r = -q_r^2$ plus rational algebraic input bounds	theorem-grade P323 lower bound	primal interval proof
O89 Oscillatory Signed Path Norm	minimum negative mass for the full phase-bearing sequence	envelope/kernel distinction	physical meaning of negativity
O90 Refreeze Stability Fiber	continuous pass/score set around QW-2038	grid nonuniqueness	out-of-sample prediction
O91 Carrier-Natural Completion Defect	transfer error of one declared map across carriers	scalar C12 overfitting	general nonscalar parent no-go
O92 Compiled Inverse Photonic Frame	mesh + short-time tomography + branch class	executable ideal transfer design	hardware realization
O93 Parallel Comb Separation Ladder	n -fold phase-product convex hull	ideal parallel discrimination	adaptive process tensor
O94 Nonparametric Clock Equivalence Tube	bounded-curvature clocks with one finite record	finite-time identifiability obstruction	external time standard
O95 Carrier Natural-ity Profile	formula versus operator invariants	prevents silent carrier transfer	categorical naturality source
O96 Typed Bridge Resource Preorder	five resource-specific free-operation pairs	formal bridge resource accounting	physical conversion laws
O97 Resource Coupling Acceptance Matrix	explicit coupling candidates and tests	blocks four shortcuts	universal coupling no-go
O98 Conditional Falsification Vector	frozen sector assumptions plus observable vector	makes historical role testable	law strict role transfer
O99 Conversion Metrology Frame	rank-three $(\ell, \tau, \hbar)$ standards	mathematics-to-units interface	emergent units
O100 External Evidence Admission Pair	hold-out and hardware contracts	prevents synthetic validation	supplies no data itself

Chapter 16

Cross-program synthesis

16.1 The most important positive result

The strict kernel is not well summarized by a single positive attenuation profile. Its phase increases the signed path resource from approximately 0.4067 to at least 0.70728. This is the clearest quantitative evidence in the round that amplitude/compression and phase/orientation are distinct mathematical resources.

16.2 The most important negative result

There is still no unique natural map

$$K_{\text{legacy}} \longmapsto K_{\text{strict}}.$$

The C12 interpolant is exact on its training carrier and catastrophically nonportable. Minimum-phase reconstruction does not recover the strict phase. The continuous refreeze point is not unique and was not hold-out validated. These facts jointly block the interpretation of the frozen strict tuple as an inevitable consequence of the legacy kernel.

16.3 The most useful missing-object construction

The most useful construction is the typed operational package

$$\mathfrak{F} = (\text{dimensionless operator, preparation, clock, instrument, environment, record, CA, sector assumptions}).$$

This is not an underlying strict object. It is the minimum explicit package needed to turn the mathematical operator into a reproducible conditional physical model. P341, P344, P348, and P349 fill parts of this package while showing exactly which inputs remain external.

Chapter 17

Recommended Research Programs P351–P364

The following order is ranked by mathematical leverage and falsifiability.

Priority	Program	Research task	Success criterion	Expected status
1	P351	Full interval-Newton primal certification of P323/P337	certified node/weight boxes, positive signs, all seven moments, matching dual value	proof-grade
2	P352	Interval certificate for the degree-11 P338 problem	global extrema of p , fixed-support weight signs, positive certified enclosure	proof-grade
3	P353	Independent QW-2038 refreeze replication	provider/registrar/analyst split, frozen hash, one unblinding, no retuning	external
4	P354	Categorical carrier-naturality theorem	graph category, morphisms, operator functor, and natural transformation or a no-go theorem	proof-grade
5	P355	Loss-aware compiled mesh digital twin	compile P341 with phase noise, insertion loss, detector POVM, and calibration inversion	strong computational
6	P356	Calibrated 12-mode photonic execution	hardware transfer matrix, raw events, controls, blinded primary endpoint	external
7	P357	Full adaptive comb SDP	common spaces, memory, instruments, causal constraints, primal/dual SDP certificate	proof/computational
8	P358	Clock design with certified nonparametric class	minimax design for $W^{2,\infty}$ or spline ball plus external clock anchor	proof/design
9	P359	Completeness of typed bridge monotones	characterize convertibility or exhibit missing monotones/catalysis	proof-grade
10	P360	Causal analytic extension theorem/no-go	prove whether any admissible analytic transfer can jointly recover strict amplitude and phase	proof-grade
11	P361	One-shot conditional electroweak blind test	preregister O98 vector, scheme, covariance, and failure rule; no post-unblinding fit	external
12	P362	Conversion-Axiom metrology package	independently calibrate $\ell_*, \tau_*, \hbar_*$ and propagate uncertainty to A	external/design
13	P363	Axiomatic FIN bridge-resource category	objects, free morphisms, tensor product, monotones, operational semantics, independence models	foundational
14	P364	Certified physical-reservoir execution	admit the P350 package and run one frozen endpoint	external

17.1 Recommended immediate sequence

$$P351 \rightarrow P352 \rightarrow P354 \rightarrow P355 \rightarrow P357$$

This sequence first closes the two most valuable mathematical certificates, then tests whether completion can be natural, and only then deepens the operational experiment.

P353, P356, P361, P362, and P364 cannot be completed by repository code alone.

Chapter 18

Reproduction

Run:

```
“text MPLCONFIGDIR=/tmp/mpl-fin python3 fin_programs_337_350.py MPLCONFIGDIR=/  
tmp/mpl-fin python3 -m unittest -v test_fin_programs_337_350.py “
```

Expected result:

```
“text Ran 18 tests OK “
```

The suite recompiles the dependency-free Lean 4.28 structural core and checks:

- the exact P337 rational feasibility margin;
- the positive P338 numerical enclosure;
- the P339 hold-out boundary;
- naturality transfer failures;
- the P341 logarithm branch theorem;
- the P343 use ladder;
- the nonparametric clock witness;
- all resource and conditional boundaries;
- all external gates;
- output-table cardinalities;
- six figures;
- JSON numerical validity.

Chapter 19

Final verdict

The present mathematical state of FIN is best described as:

a dimensionless, phase-bearing spectral graph operator with two dual dynamical semigroups, a quantitatively irreducible signed path resource, and a growing operational resource theory — but without a natural legacy-to-strict source map, a non-premise selector, an internal unit system, or independent physical validation.

The deepest advance of P337–P350 is not a new physical claim. It is a sharper factorization of what a physical claim would require:

operator shape + signed/phase resources + carrier-natural source law
+ clock/instrument/record + conversion standards
+ sector assumptions + independent evidence.

No current theorem compresses these components into one unavoidable strict object. P346 shows how to organize them; P347 shows that four tempting shortcuts fail; P348–P349 show how to proceed honestly on a conditional physics branch. This is a scientifically useful result because it converts ambiguous "missing physics" into a finite, typed, falsifiable program.

Chapter 20

References and repository sources

1. FIN repository guardrails: AGENTS.md.
2. Kernel split and provenance packets: K1, K2, F2, F3, and S2.
3. SUMMARY_GROK.md.
4. QW-2038 derivation-compatible refreeze source and JSON report.
5. FIN Releases 10.27–10.29 and Programs P295–P336.
6. Generated numerical files and the formal source accompanying this report.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/mpl-fin \  
python3 fin_programs_337_350.py
```

```
MPLCONFIGDIR=/tmp/mpl-fin \  
python3 -m unittest -v test_fin_programs_337_350.py
```

Expected result: 18 tests passed, 0 failed. The suite recompiles the dependency-free Lean 4.28 structural core. Refreeze search, photonic calibration, detector models, clocks, and metrology designs retain the boundaries stated in the text. P342 and P350 admit no external evidence.

Suggested citation:

Żuchowski, K. (2026). *FIN Research Programs P337–P350: Certified Signed Resources, Carrier Naturality, Operational Compilation, and Conditional Physics Interfaces* (Release 10.30; Version 1.0.0) [Preprint]. Zenodo.